Mukta Choudhury

GENAI SECURITY | APPLICATION SECURITY | SECURITY AUTOMATION

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LinkedIn | GitHub | Portfolio

Profile

AI security and compliance analyst working at the intersection of GenAI SaaS, agentic AI and secure software development. Brings an investigative mindset to understanding how systems behave, where trust breaks down and which controls reduce real risk. Combines hands-on engineering with adversarial testing, privacy awareness and clear communication. Comfortable moving between technical detail and business impact to help teams build, evaluate and adopt AI with confidence.

Experience

Adtecher | AI Security and Compliance Analyst | Feb 2026 - Present

- Assess application and API security in a GenAI SaaS environment, covering tenant isolation, access control, secrets and cloud configuration; turn technical findings into prioritised remediation and traceable control evidence.
- Evaluate agentic AI workflows through adversarial testing for prompt injection, sensitive-context disclosure, unsafe output rendering and approval bypass; completed 19 authenticated checks with reproducible evidence and clear finding criteria.
- Uncovered an agent approval-control failure through end-to-end action verification, exposing a gap between user intent, execution and audit records; supplied reproduction evidence and actionable requirements for human-in-the-loop controls.
- Developed ComplianceHub from a personal project into an internal tool used at Adtecher to track security controls, accountable owners, evidence and remediation in one place.
- Validate integration resilience through controlled GitHub access-loss, recovery and replay scenarios; trace notification failures across provider events, application state and delivery queues.

Future Interns | Cyber Security Intern | Nov 2025 - Dec 2025

- Performed practical web-application security assessments with Burp Suite, OWASP ZAP and Nikto, translating SQL injection, XSS and access-control findings into clear remediation guidance.
- Analysed 1,000+ authentication and network events in Splunk for suspicious activity; developed an AES-encrypted Flask file-handling module to practise defensive application design.

Vehere | Pre-Sales Intern | Mar 2025 - May 2025

- Supported customer-facing IDS/IPS evaluations by configuring demonstration environments, explaining detection behaviour and translating customer requirements into documented technical actions.

University of Sheffield | Student Support Worker | Sep 2024 - Jun 2025

- Resolved accessibility and technical support needs, coordinated IT escalations and made technical guidance accessible to non-technical users.

Technical skills

GenAI and application security: adversarial testing, prompt injection, human-in-the-loop controls, guardrail evaluation, API security, privacy review and tenant isolation.

Engineering and platforms: Python, TypeScript, Next.js, FastAPI, Flask, PostgreSQL, Docker, GitHub Actions, AWS development workflows and AI-assisted development.

Security tools and assurance: Burp Suite, OWASP ZAP, Nikto, Splunk, Prowler integration, CVSS, EPSS, CISA KEV, STRIDE, risk registers and ISO 27001 readiness.

Selected projects

ComplianceHub | Internal governance and ISO 27001 readiness platform

- Built a central compliance workspace that connects risks, policies, assets and control evidence, giving the team a structured way to manage ISO 27001 readiness and follow through on security responsibilities.
- Translated compliance requirements into plain-English assessments, a versioned Statement of Applicability and a 5x5 risk register, turning identified gaps into prioritised treatment plans and assigned tasks.
- Implemented policy approvals, employee acknowledgements and evidence-review workflows, linking records to controls, owners and review dates so compliance activity remains traceable as the business grows.
- Built an internal-audit workflow covering audit planning, control testing, findings and corrective actions, with exportable evidence packs and leadership reports to support audit preparation and management review.
- Integrated GitHub and cloud security monitoring with Slack alerts and linked remediation tasks, connecting technical findings to the team's compliance follow-up and security oversight.

Cybertrend Intelligence | Actionable threat intelligence and vulnerability prioritisation

- Built a Python threat-intelligence pipeline that brings security reporting and vulnerability data into one prioritised feed, helping focus attention on threats that warrant action.
- Combined CVSS severity, EPSS exploitation probability and CISA KEV evidence with structured GenAI summaries to explain affected assets, business risk and recommended responses.
- Automated severity-grouped email briefings and critical alerts, with deduplication and stored delivery records to keep intelligence actionable and traceable.

Privacy Review Agent | Privacy controls embedded in software delivery

- Built an offline-first review tool that flags exposed secrets, personal-data logging and sensitive information passed to LLMs, bringing privacy checks into commits and pull requests.
- Turned policy rules into severity- and confidence-based CI checks, with remediation guidance, reviewable exceptions and standardised reports to help developers address risks before merging code.

AI security testing | Adversarial evaluation and risk reporting

- Created a repeatable local test framework for direct and indirect prompt injection, jailbreaks and tool-action hijacking, using a mock application to compare stronger and weaker guardrails.
- Translated test outcomes into risk registers and mitigation priorities, linking likelihood, impact and control effectiveness to OWASP LLM risks and ISO 27001 control themes.

Threat modelling | Security analysis for connected medical devices

- Modelled an RFID-enabled medical-device ecosystem using STRIDE, identifying how spoofing, data tampering and service disruption could affect patient safety and privacy; proposed authentication, access-control and audit safeguards.

Education

University of Sheffield | PG Diploma in Cyber Security and Artificial Intelligence | Jan 2026

Amity University | Bachelor of Computer Applications | Mar 2023

Certifications and training

Azure Fundamentals (AZ-900) training completed
TryHackMe Pre Security and SOC Level 1 learning paths | AIG Shields Up Cybersecurity Job Simulation
CompTIA Security+ | Preparation in progress; target exam first week of November 2026
AWS Certified Cloud Practitioner | Preparation in progress; target exam first week of November 2026